Project 2: Overview, Research Standards, and Grading

CIS 6270: Discrete Generative Models

Fall 2026

Project assigned: Thursday, October 22.
ICML-ready paper outline due: Tuesday, November 10, at 8:30 am.
Complete paper and code due: Thursday, December 3, at 8:30 am on Canvas.
Project defenses and slides due: Friday, December 4.
Paper format: Anonymous ICML *main-conference submission* format, with references and an appendix. Until the ICML 2027 rules are released, use the ICML 2026 submission standard as the provisional reference: at most 8 pages of main text, excluding references and appendices.
Total project score: 40 points.

Document Outline

Click a heading below to jump to that section.

- Project Overview
- Grading Summary
- A. Research Outline
- B. Research Paper & Code
  - B1. Novelty and Research Significance
  - B2. Technical Formulation and Soundness
  - B3. Experimental Rigor and Scientific Evidence
  - B4. Scientific Writing, Figures, and Formatting
  - B5. Code and Reproducibility
  - Paper Organization and Scientific Writing Standards
- C. Final Defense
- Scoring and Partial-Credit Policy
- Words of Advice

Project Overview

In this project, you will **develop and evaluate a novel discrete generative formulation** and write a complete research paper that meets the standards of an **ICML main-conference submission** (paper template provided as a separate PDF; please review). Strong projects may be considered for submission to ICML 2027.

Research scope. The key contribution must concern learning, representing, or sampling generative distributions over *discrete objects*.

- **Modalities** include, but are not limited to: categorical or biological sequences, language and code, graphs, discrete structures and events, and variable-length objects.
- **Model choices** include, but are not limited to: discrete diffusion, masked models, discrete flow matching, continuous-time Markov chains, jump processes, and related discrete formulations.
  - Methods based on probability-simplex relaxations, continuous latent variables, or hybrid continuous–discrete dynamics are allowed, provided that the innovation makes a substantive contribution to discrete generation. Define the connection between the continuous representation and the generated discrete

objects, including any decoding or discretization step. For a continuous generator with a discrete decoder, identify the new mechanism that contributes to discrete generation.

- **Study design**:
  - **Required:** well-defined problem statement, methodological contribution, relevant benchmarks, strong competitive baselines, controlled ablations, quantitative evaluation with multiple independent experimental runs, an honest analysis of results, complete references, an appendix, and a reproducible code repository. Support methodological originality through a technical comparison with prior work, and assess empirical performance through experiments.
  - **Team's choice:** number of modalities (one may be sufficient, if the analysis is thorough) and the specifics of the datasets, external baselines, seeds, and ablations. Guidance, cross-modality transfer, and new theorems are optional contributions.
- **Standard for methodological novelty:** a new mechanism or formulation with clear technical contribution.
  - **NOT valid:** dataset selection, random seeds, optimizer settings, training duration, model size, backbone choice, method names, and standard guidance or loss terms. These are all important elements of your study, but they do *not* count as the methodological contribution.
  - A combination of published components qualifies as novel when the team develops and explains a substantive new formulation.

Grading Summary (40 points)

Deliverable	Due date	Points
A. Research Outline	November 10	5.0
B. Research Paper & Code	December 3	15.0
C. Final Defense	December 4	20.0
Total		**40.0**

Individual grades will be normalized by CATME. Please answer your CATME surveys honestly.

A. Research Outline — 5 points

Upload for the Research Outline

1. **ICML-formatted paper outline.** Include title, abstract, introduction, related work, preliminaries, proposed formulation, planned experiments and results, discussion/limitations, references, and a planned appendix. Use labeled placeholders for pending results; distinguish hypotheses from completed findings.

A1. Research Problem, Gap, and Novelty (1.0 pts)

- Define the discrete generation task, intended capability, and the specific problem to solve. **(.25 pt)**

- Identify and accurately describe the closest prior work and its relevant limitation; cite the evidence for the gap. **(.25 pt)**
- State the proposed new idea, what differs from prior work, and the central testable research hypothesis. **(.5 pt)**

A2. Proposed Formulation **(1.5 pts)**

- Specify the state space, inputs/outputs, distributions, conditioning, and the relationship to discrete samples. **(.5 pt)**
- Define the core mathematical or algorithmic mechanism beyond a purely conceptual description. **(.5 pt)**
- Describe the intended training objective and inference/sampling procedure at sufficient detail to assess feasibility. **(.5 pt)**

A3. Experimental Plan and Claim–Experiment Mapping **(1.5 pts)**

- Choose relevant datasets/benchmarks and metrics, with an intended evaluation protocol. **(.5 pt)**
- Identify strong baselines, closest-method comparisons, and the main fairness controls. **(.5 pt)**
- Plan informative ablations, independent runs and uncertainty reporting, and any analyses needed for the principal claims. State which proposed experiment tests each claim. **(.5 pt)**

A4. Complete ICML-Style Paper Outline **(1.0 pts)**

- Include the sections listed in the outline submission requirements, with labeled placeholders in planned results tables. **(.5 pt)**
- Present an organized, readable manuscript with consistent terminology and enough explanations to follow the proposal. **(.25 pt)**
- Provide appropriate references and a planned appendix covering proofs, implementation details, and supplemental experiments as applicable. **(.25 pt)**

Changes after the outline. You may revise the plan; the outline grade remains based on the submitted version.

B. Research Paper & Code — 15 points

Upload to Canvas for the Research Paper & Code

1. **Complete ICML-style research paper** with results, references, and an appendix.
2. **Manuscript source files** needed to compile the paper.
3. **Full code repository** with instructions to reproduce the main experiments, tables, and figures.

B1. Novelty and Research Significance 3.5 points

B1.1. Research Gap and Positioning Against Prior Work (.75 pts)

- Motivate a specific limitation, opportunity, or unmet capability in discrete generation. **(.25 pt)**
- Accurately identify and discuss the most technically relevant work, especially the closest published or publicly available approaches. **(.25 pt)**
- Explain which parts are already established and where the proposed contribution differs. **(.25 pt)**

B1.2. Substantive Methodological Novelty (1.5 pts)

- Introduce an identifiable, nontrivial contribution to the formulation or mechanism of discrete generative modeling, with an explicit change to the generative mechanism. **(.5 pt)**
- Make the difference from the closest prior methods technically explicit and correctly attribute reused components. **(.5 pt)**
- Show why the contribution is specific to discrete generation, or substantively changes how discrete samples are modeled or generated, including for simplex/latent/hybrid approaches. **(.5 pt)**

B1.3. Research Hypothesis and Mechanistic Motivation (.75 pts)

- State a clear hypothesis linking the identified limitation to the proposed mechanism. **(.25 pt)**
- Explain why the new design is expected to work using mathematical, probabilistic, computational, or task-specific reasoning. **(.25 pt)**
- Make the mechanism testable through specific ablations, comparisons, or observable predictions. **(.25 pt)**

B1.4. Research Significance and Appropriate Contribution Claims (.5 pts)

- Explain why the contribution matters for a modeling capability, quality, efficiency, control, generality, or theoretical question. **(.25 pt)**
- State the contribution within the task, data, and conditions evaluated. **(.25 pt)**

Novelty statement. In the Introduction, Related Work, or Methods, identify the closest prior formulation, what is reused, what is new, and why it matters.

B2. Technical Formulation and Soundness 3.0 points

B2.1. Problem, State Space, and Notation (.5 pts)

- Define the discrete objects/state space, source/target distributions, relevant time conventions, and generation task. **(.25 pt)**
- Define conditioning and structural constraints, any simplex/latent/decoding or variable-length mapping, and all mathematical symbols unambiguously and consistently. **(.25 pt)**

B2.2. Mathematical and Probabilistic Correctness (1.0 pts)

- **Validity of Mathematical Objects.** Ensure that all defined probabilistic objects and generative mechanisms are mathematically valid, satisfying applicable normalization, nonnegativity, transition-rate or kernel, and state-space constraints. **(.5 pt)**
- **Consistency of the Formulation.** Establish mathematically correct relationships among the proposed probability paths, dynamics, prediction targets, and training objectives, ensuring that they jointly define a coherent generative process. **(.25 pt)**
- **Assumptions and Theoretical Claims.** Ensure that stated assumptions, equations, and claimed mathematical properties are mutually consistent, and distinguish exact results from approximations and heuristics. **(.25 pt)**

Formulation checks. Check rate validity for jump processes, corruption/reverse-sampling consistency for diffusion, path/rate agreement for flow matching, and which properties survive decoding for continuous representations.

B2.3. Complete Model and Algorithm Specification (.75 pts)

- Describe the architecture, parameters, model components, and complete training objective/procedure. **(.25 pt)**
- Describe inference and sampling in enough detail to determine how discrete outputs are obtained. **(.25 pt)**
- Give reproducible algorithmic detail: essential schedules, update rules, component interfaces, and trainable/pretrained/frozen distinctions where relevant. **(.25 pt)**

B2.4. Derivations, Guarantees, and Technical Justification (.5 pts)

- Justify new formulas or algorithmic choices and provide correct derivations or references for nontrivial identities used. **(.25 pt)**
- Support any claimed theorem or guarantee with a valid proof and assumptions; otherwise distinguish a heuristic or empirical observation from an established fact. **(.25 pt)**

B2.5. Assumptions, Constraints, and Computational Behavior (.25 pts)

- State modeling assumptions, restrictions, and relevant computational costs, bottlenecks, or complexity trade-offs. **(.25 pt)**

B3. Experimental Rigor and Scientific Evidence 5.0 points

B3.1. Benchmarks, Metrics, and Evaluation Protocol (.75 pts)

- Identify data sources, versions, sample counts, filtering, train/validation/test split, and leakage controls. **(.25 pt)**
- Choose metrics aligned with the proposed task and claims; define directions, units, implementation, and relevant limitations. **(.25 pt)**
- State conditioning, generated-sample counts, checkpoint selection, evaluation scripts, and any task-specific protocol clearly. **(.25 pt)**

B3.2. Competitive Baselines and Fairness **(1.0 pts)**

- Compare with the closest prior approach, an appropriate internal/base-model control, and strong external competitors as relevant to the task; justify omitted major comparators. **(.5 pt)**
- Use comparable tasks, data, conditions, metrics, and reasonable compute/sampling or tuning budgets whenever possible. **(.25 pt)**
- Distinguish independently reproduced, reimplemented, and literature-reported results; disclose material differences, missing baselines, and resource limitations. **(.25 pt)**

Baseline selection. Choose the number of baselines according to their relevance and competitiveness. Include the closest and strongest comparators appropriate to the task. State differences in comparison conditions and account for them in the interpretation.

B3.3. Main Results, Statistical Reliability, and Empirical Improvement **(1.25 pts)**

- Report complete main quantitative results, including appropriate quality and trade-off metrics and unfavorable as well as favorable outcomes. **(.25 pt)**
- Use justified independent random seeds or experimental repetitions; distinguish training variation from repeated sampling of one checkpoint and report appropriate uncertainty. **(.25 pt)**
- Demonstrate measured improvement over relevant baselines for the proposed task and claims, supported by the metrics needed to evaluate the claim. **(.75 pt)**

Independent runs. Choose and justify the number and type of repetitions needed for a credible study. Define what the error bars and summary statistics measure. Distinguish independent training runs from repeated sampling of one checkpoint. For sampling-only or prohibitively expensive methods, justify the repetition protocol and describe its statistical scope.

Negative or mixed results. Report and analyze the observed outcome of each experiment. Complete reporting and reliable statistics earn credit under the corresponding B3.3 criteria, and research rigor earns credit under the other applicable criteria. The 0.75 improvement points require convincing evidence of an empirical advantage relevant to the task.

B3.4. Controlled Ablation Studies **(1.0 pts)**

- Compare the proposed method against a defined base model or control. **(.5 pt)**
- Test the effect of new components or design choices by removing or replacing them while holding other conditions fixed. If a component cannot be removed without invalidating the method, compare against a valid alternative that tests the same design choice. **(.25 pt)**
- Control major confounders such as parameter count, sampling budget, auxiliary predictors, or extra computation where they could explain the reported gain. **(.25 pt)**

Each ablation should test an explanation for the observed behavior. Hold other relevant factors fixed when removing or replacing a component. Choose the number of ablations to cover the design choices and claims of the method.

B3.5. Sensitivity, Robustness, and Failure Analysis **(.5 pts)**

- Investigate relevant sensitivity, stability, robustness, scalability, or generalization questions, especially those created by new hyperparameters or assumptions. **(.25 pt)**

- Identify and analyze limitations, failure modes, or trade-offs; support the analysis with experimental evidence. **(.25 pt)**

B3.6. Interpretation and Claim–Evidence Consistency (.5 pts)

- Connect each main claim to the exact experiment that tests it, distinguishing direct observations from interpretations or speculative explanations. **(.25 pt)**
- Ensure consistency of all numerical values, metric definitions, settings, and conclusions across the paper and appendix, and interpret favorable and unfavorable results within the scope of the evidence. **(.25 pt)**

Evaluation bias. If a learned predictor is used in training or guidance, use independent evaluation where feasible and disclose remaining dependence. Include auxiliary computation in efficiency claims.

B4. Scientific Writing, Figures, and Formatting 2.0 points

B4.1. Organization, Scientific Narrative, and Writing (.5 pts)

- Present a coherent story from research problem and prior gap through mechanism, experiments, interpretation, and conclusion. **(.25 pt)**
- Use precise, concise, readable academic prose; define terms, focus the background on the research question and support performance claims with evidence, and explain why each result matters. **(.25 pt)**

B4.2. Professional Figures and Tables (.75 pts)

- Include an original, scientific *method overview* figure that illustrates the proposed mechanism and distinguishes the key innovation from standard components. **(.25 pt)**
- Make all figures publication-quality: readable at two-column scale, with accurate labels, axes, units, legends, panel identifiers, and informative captions *below* the figures; reference and discuss them in the text. **(.25 pt)**
- Format tables professionally using the `booktabs` style (`\toprule`, `\midrule`, `\bottomrule`), captions *above* tables, consistent precision, metric direction/units, explained uncertainty, and correct in-text references. **(.25 pt)**

B4.3. Mathematical Notation and Technical Terminology (.25 pts)

- Define mathematical symbols, domains, and technical terminology consistently; explain the equations in connected prose. **(.25 pt)**

B4.4. ICML Format, References, and Appendix (.5 pts)

- Use the appropriate anonymous ICML submission template, obey the current main-body page limit and layout rules, and preserve the official style, margins, and fonts. **(.25 pt)**
- Cite contributions, data, figures, code, and baselines accurately and consistently. Include essential evidence in the main paper and supporting details in the appendix. **(.25 pt)**

B5. Code and Reproducibility 1.5 points

B5.1. Repository Structure, Setup, and Execution (.25 pts)

- Provide a clear README, repository structure, environment and dataset instructions, and runnable commands for training, sampling, evaluation, and regeneration of the principal results. **(.25 pt)**

B5.2. Implementation Correctness and Paper Consistency (.75 pts)

- Correctly implement the actual proposed mechanism, including the applicable probability, loss, transition, and sampling operations. **(.5 pt)**
- Ensure implementation matches the paper's equations and stated settings; include relevant smoke tests or correctness checks for nontrivial components. **(.25 pt)**

B5.3. Experimental Traceability and Reproducibility (.25 pts)

- Provide configurations, seeds, checkpoints, available logs/raw metrics, and the scripts or data used to produce the main tables and figures so results can be traced to actual runs. **(.25 pt)**

B5.4. Documentation and Attribution (.25 pts)

- Document non-obvious implementation choices and execution restrictions, and attribute borrowed repositories, pretrained models, benchmarks, and data. **(.25 pt)**

Reproducibility. Graders may assess expensive experiments through saved outputs and targeted runs.

Paper Organization and Scientific Writing Standards

Sections may be combined or renamed. Keep the method definitions, central comparisons, and innovation ablations in the main text; place full hyperparameters, proofs, and supplemental results in the appendix. Use the separate paper template for organization.

Figures and tables. Follow B4.2. Use LaTeX captions, labels, and cross-references. Prefer vector graphics; use accessible colors or distinct line styles. Explain normalization, error bars, aggregation, smoothing, table symbols, and comparison conditions. Avoid misleading axis ranges or example selection.

Integrity. Verify sources and ensure each citation supports its claim. Report failed runs and unfavorable results that affect conclusions; disclose pretrained components, external evaluators, and data exclusions. Teams are responsible for all submitted content, including AI-assisted work, and must follow course, university, and conference policies on its use and disclosure.

C. Final Defense 20 points

Defense requirements

1. **Upload presentation slides to Canvas.**

Timing: 20-minute presentation (hard limit) + 10-minute Q&A.

Slide labels: Cover every required ID in Parts 1–5 on a dedicated slide or sequence. Put **1A, 1B, etc. in the top-right corner**. Follow the general order below; reorder when needed for clarity and repeat the ID on continuation slides. Title, transition, and backup slides are allowed.

Citations: Provide credit for intellectual contributions and images using [1] or superscripts. Put brief matching sources in small, readable text at the **bottom right**; numbering may restart on each slide. Identify original figures. End with a **Bibliography slide** in a consistent style; reference order is flexible.

Presentation Content 18 points

Points are earned through clear, accurate oral explanations during the presentation. Q&A can revise any item's score, including reducing full credit to zero when answers reveal knowledge gaps. There is no separate Q&A score; each item stays within its stated point range.

Every member must understand the whole project and answer all questions directed individually to them. The shared defense score is subject to the CATME normalization described above.

Part 1: Research Question and Motivation (2.0 pts)

- **Slide 1A:** Define the discrete-generation problem, task setting, and motivation clearly. **(1.0 pt)**
- **Slide 1B:** Identify the concrete gap or limitation that the project aims to address and state the research question or hypothesis. **(1.0 pt)**

Part 2: Novelty and Positioning Against Prior Work (3.0 pts)

- **Slide 2A:** Identify the most closely related formulations and explain their relevant limitations accurately, with appropriate citations. **(1.5 pt)**
- **Slide 2B:** Distinguish the team's original formulation from reused components; explain why the new mechanism is substantive and discrete-aware. **(1.5 pt)**

Part 3: Method and Mathematical Formulation (6.0 pts)

- **Slide 3A:** Present an informative method-overview figure and explain how the principal innovation changes the generative mechanism. **(1.5 pt)**
- **Slide 3B:** Define the state space, distributions, and conditioning. Explain the core equations or update rules, relevant constraints, assumptions, approximations, and the scope of any claimed guarantees. **(2.5 pt)**
- **Slide 3C:** Explain the implemented architecture, training objective and procedure, and sampling procedure, including how valid discrete outputs are obtained. Connect the formulation to implementation and explain the principal design choices. **(2.0 pt)**

Part 4: Experimental Evidence and Research Findings **(5.0 pts)**

- **Slide 4A:** Present the central benchmark comparisons with clear metrics, competitive baselines, uncertainty reporting, and fair or disclosed comparison conditions. Explain the choice of benchmarks, data splits, metrics, and comparison budgets. **(2.0 pt)**
- **Slide 4B:** Present informative controlled ablations that isolate the core innovation, and explain which hypotheses they test and which alternative explanations they address. **(1.5 pt)**
- **Slide 4C:** Accurately interpret the main quantitative results, independent runs, and uncertainty, including trade-offs and cases that are neutral, unfavorable, or inconclusive. Explain whether the evidence supports the stated claims. **(1.5 pt)**

Part 5: Limitations and Conclusions **(2.0 pts)**

- **Slide 5A:** State the main conclusion and explain how the results support it. **(1.0 pt)**
- **Slide 5B:** Explain a substantive limitation or failure mode and a compelling follow-up experiment, using the underlying modeling principles to justify what it would test. **(1.0 pt)**

Overall Presentation Quality **2 points**

- **Timing:** Presentation covers the full study within the 20-minute limit; the defense then allows 10 minutes for Q&A. **(0.7 pt)**
- **Quality:** Well-rehearsed presentation with clear explanations, readable figures, and smooth transitions. **(0.7 pt)**
- **Even contribution:** All team members contribute evenly to the presentation and participate in the defense. **(0.6 pt)**
- General presentation quality penalty (up to -0.3 points; document a distinct reason for the deduction). **(0 pt)**

Words of Advice

- Choose a project you can complete with the available time and compute.
- Verify the baseline implementation before adding your method.
- Write while experimenting; update the paper as results arrive.